Pawel Balawender

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Work Experience

Sentinel Eagle
Software Developer, Warsaw — October 2025 – July 2026

- New startup founded by Marek Cygan. Building non-GPS drone navigation for the war in Ukraine.
- One of the first employees.
- My part: kernel / OS works; drivers for hardware; hardware security.
- Manual assembly of dozens of first devices.

Education

University of Warsaw (MSc: 5.0 / 5.0) — October 2023 – December 2025

- Discovered that Lean 4 enables a new way to formalize time complexity of programs.
- Results presented at AITP'25 in France: ruplet.com/aitp-presentation.pdf
- Research visit to INRIA Paris with Yannick Forster.
- Top 5 students selected for special research funding.

University of Warsaw (BSc: 4.5 / 5.0) — October 2020 – July 2023

- Analyzing how effectively LLMs influence discourse on Twitter, with a focus on hate speech against Ukraine.
- Fine-tuning GPT-2 for Polish text generation.
- 20 citations as of August 2026; part of an ERC-funded research led by Piotr Sankowski.
- Presented at EMNLP 2024 in Miami (Findings). Paper: arxiv.org/abs/2311.16905
- Top 8 bachelor's students selected for a special course of study during the sophomore year.

Selected Experience

Optiver
Software Engineering Intern, Amsterdam — July 2024 – August 2024

- Working in low-latency C codebase as part of the Automated Trading Systems team.
- Solving a difficult problem in thread synchronization and shared memory utilization.
- €20,000-per-month internship; chose to not pursue career in finance.

Goldman Sachs
Summer Analyst, Warsaw — July 2022 – September 2022

- Deploying a monitoring application in Python using Docker, Kubernetes, and Terraform.

MIM Solutions
Junior ML Engineer — October 2020 – September 2021

- Working half-time during my first year at university.
- Designing a newspaper-digitization system using classical NLP and ML.
- Fine-tuning BERT models for classification.

Award from the Polish Ministry of Education — July 2020

- Received a Ministry award for the most accomplished high school students of the year.

European Union Contest for Young Scientists — October 2019 – June 2020

- In a team of two, designed a system for solar charging UAV batteries mid-flight.
- Developed bare-metal drivers in C and Assembly for an STM32 microcontroller.

Dulwich College London — September 2019 – June 2020

- Selected for funding + accommodation to study A-levels at Dulwich College London.
- Result of support from the Polish Children's Fund throughout high school (2017–2020).

Operating System for the Cortex-M0 microprocessor — November 2018 – March 2019

- Won 1st place in Poland in the MeetIT Foundation's competition.
- Designed a minimal operating system with threading and a file system.